

Demo: Homopolar Motor

Contact, force, reversal, and heat from timed evidence

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

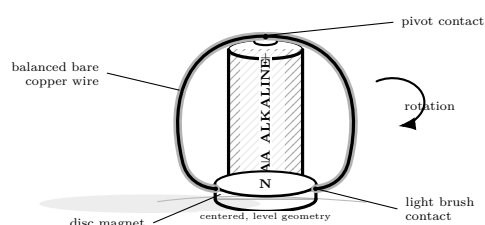
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- One fresh AA alkaline cell
- One neodymium disc magnet
- Bare copper wire
- Pliers
- Timer
- Paper direction flag
- Room-temperature reference cell

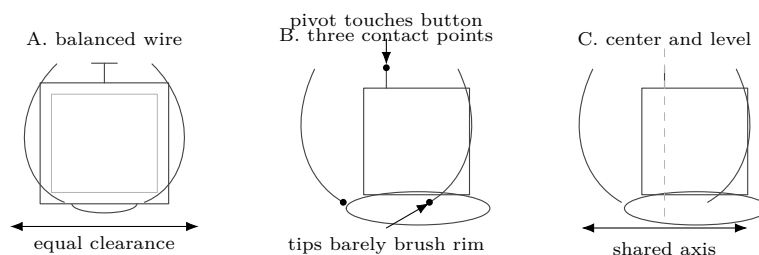
Predict first

First sketch the expected view from above. If the magnet flips, should rotation reverse? If both cell and magnet flip, should it reverse or match the first trial?



Investigate

1. Shape and inspect — no power

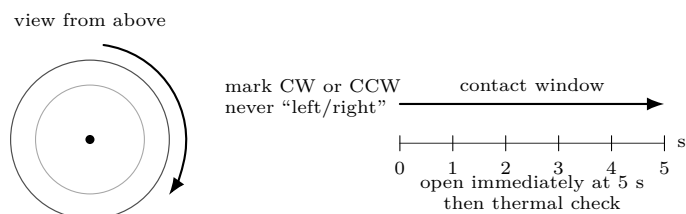


1. Adult counts one magnet, checks that it is unchipped, and mounts it on the negative end. Learner bends the wire while it is away from the cell. Adult verifies symmetry, clearance, and three light contacts.
2. Learner answers before power: Which contact completes the circuit? Where can friction steal motion? Which parts may heat first?

2. Time one baseline trial

3. Put a paper direction flag behind the apparatus. Adult clears loose metal and starts the timer, then lowers the wire. Call “contact” at 0 s; learner counts full turns and calls clockwise or counterclockwise from above.

- Adult lifts the wire at 5 s or sooner for stall, odor, pain, sparking, wobble, or unexpected motion. Record start delay, direction, and turns in five seconds; do not extrapolate a speed from a partial turn.
- With power open, adult compares wire, cell, and magnet with the room-temperature reference using the back of a finger. If any is distinctly warm, damaged, or uncomfortable, stop for the day. Otherwise cool at least 60 s before another trial.



3. Test reversals

- Predict every row before testing. Adult tests only rows needed to distinguish the rule, with the same wire geometry, five-second limit, stop line, and cool-down.

Trial	Cell orientation	Magnet face down	Prediction	Observed
A	+ at pivot	N	baseline: _____	_____
B	+ at pivot	S	reverse / same	_____
C	– at pivot	N	reverse / same	_____
D	– at pivot	S	reverse / same	_____

Adult setup note: reversing the cell requires remounting the magnet on the opposite terminal and reshaping the wire; never force an unstable stack. Skip C and D if the magnet will not sit securely.

4. Diagnose without extending contact

Symptom	Inspect with circuit open	One correction
No motion	Missing pivot or rim contact; coating on wire	Clean bare contact points; rebalance
Starts only when pushed	Excess friction; tips pressing inward	Open tips slightly; restore symmetry
Strong wobble	Pivot off-axis; unequal arms; tilted magnet	Recenter and level before retry
Sparks or rapid heat	Contact pressure, damaged cell, short path	Stop; retire damaged parts; no retry
Direction changes mid-trial	Intermittent contact or rocking magnet	Open circuit and rebuild geometry

Explain from the evidence

Current crossing the magnet's field experiences force; the opposed wire arms turn that force into torque about the cell axis. Reversing current or magnetic field reverses the force and rotation. Reversing both preserves the direction. Contact quality changes whether torque beats friction, but it does not change the reversal rule. This apparatus is nearly a direct short, so much of the input can become heat. That is why every energization is timed, adult-controlled, AA alkaline only, never lithium, and separated by an open-circuit thermal check.

Change one thing

Final proof: Use the smallest safe set of trials to support the claim "one reversal reverses motion; two reversals restore it." Name the two trials compared, report direction from above and turns in five seconds, then identify one result that would

have falsified the claim. Finish only when the adult confirms the circuit is open, all parts are cool and undamaged, the single magnet is counted into storage, and no wire remains touching the cell.

Evidence record	
Prediction	_____
One change	_____
Observation	_____
Claim + because	_____

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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